Interferometric Circuits Design, Simulation, Layout Design, and Validation

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***Abstract*— The project aims to create an interferometric circuit from which the waveguide's group index can be extracted, in accordance with the specifications set out in the "Layout/Report Submission" task of the Silicon Photonics Design, Fabrication and Data Analysis course taught by Professors Lukas Chrostowski and Mateo Branion-Calles via edX.**

Keywords—group index, interferomer, photonics, wageguide

I. INTRODUCTION

II. THEORETICAL FRAMEWORK

A. Waveguide

An optical waveguide is a linear physical structure composed of transparent dielectric materials, designed to confine, guide, and channel light (in the visible, infrared, or ultraviolet spectrum) along a specific path. Unlike metallic microwave waveguides, light confinement occurs due to the geometric transition between materials with different refractive indices, relying primarily on the phenomenon of total internal reflection or interference effects in photonic crystals.

1) Effective Refractive Index - n_{eff}

The effective refractive index is a fundamental dimensionless parameter in waveguide optics and integrated photonics. It quantifies the overall phase velocity at which a specific optical mode propagates along the waveguide structure.

Unlike the refractive index of a bulk material—which is an intrinsic property of matter—neff is a structural property of the mode. It depends simultaneously on the waveguide geometry, the light wavelengths, and the refractive indices of all regions making up the structure (such as the core and cladding).

The linear propagation constant (β) of an optical mode describes how the wave's phase varies along the direction of propagation (z). The effective refractive index is formally defined as the ratio between this propagation constant and the vacuum wavenumber (k0):

$$n_{eff} = \frac{\beta}{k_0} = \frac{\beta\, c}{\omega} \qquad (1)$$

Where:

β is the mode propagation constant (rad/m).

$k_0 = 2\pi/\lambda_0$ is the vacuum wavenumber.

c is the speed of light in a vacuum.

ω is the angular frequency of the light. The phase velocity (vf) of the mode within the waveguide is inversely proportional to neff:

$$v_p = \frac{c}{n_{eff}} \qquad (2)$$

2) Physical Boundaries and Confinement

For a guided mode (which does not lose energy via radiation outside the structure), the value of neff is strictly confined between the refractive indices of the materials forming the waveguide. In a waveguide composed of a core (ncore) and a cladding (ncladding):

$$n_{cladding} < n_{eff} < n_{core}$$

When close to ncore: This means the mode is highly confined within the core. Most of its electromagnetic energy travels through the central material.

When close to ncladding: This means the mode is near its cutoff limit. The evanescent field extends deeply into the cladding, making the mode sensitive to external perturbations.

If $n_{eff} < n_{cladding}$, the mathematical solution ceases to represent a guided mode and instead represents a radiation mode (light escapes the waveguide).

n_{eff} is a key variable in the design of modern optical components:

Phase Matching: Essential for designing directional couplers, Bragg gratings, and electro-optic modulators.

Waveguide Dispersion: The variation of n_{eff} with wavelength (λ) alters the group velocity of light, generating chromatic dispersion that must be controlled in telecommunications.

3) Effective Area (Aeff)

The effective area is a crucial geometric-optical parameter in the design of photonic waveguides and optical fibers. It quantifies the actual cross-sectional area over which the optical mode power is distributed as it propagates through the structure.

Unlike the geometric physical area of the waveguide core, the effective area takes into account the spatial distribution of the electric field intensity, including the portion of light traveling outside the core as evanescent fields.

Mathematically, the effective area of a propagating mode is calculated from the two-dimensional surface integral of the transverse electric field profile (E(x,y)). For waveguides composed of isotropic materials, the standard formula is defined as:

$$A_{eff} = \frac{\left(\iint_{-\infty}^{+\infty} |E(x,y)|^2\, dx\, dy\right)^2}{\iint_{-\infty}^{+\infty} |E(x,y)|^4\, dx\, dy} \qquad (3)$$

Where:

- E(x, y) is the transverse electric field amplitude distribution of the mode.
- The integrals cover the entire infinite transverse space (x, y), encompassing both the core and the cladding.

If the mode profile is approximated by an ideal Gaussian distribution with a mode field radius (ω), the effective area can be simplified to:

$$A_{eff} = \pi\omega^2 \quad (4)$$

The effective area is the parameter that determines the optical power density within the waveguide. Its primary application lies in controlling nonlinear optical effects (such as the Kerr effect, Stimulated Raman Scattering—SRS, and Four-Wave Mixing—FWM). The nonlinear efficiency of a waveguide is inversely proportional to its effective area and is governed by the nonlinearity coefficient (γ):

$$\gamma = \frac{2\pi n_2}{\lambda A_{eff}} \quad (5)$$

Where:
n_2 is the material's nonlinear refractive index and λ is the wavelength.

Low A_{eff} (Integrated Photonics): In waveguides based on silicon or silicon nitride (SiN) nanowires, strong optical confinement reduces the A_{eff} to fractions of a square micrometer (< 1 μm²). This maximizes nonlinear effects, enabling the creation of frequency combs and supercontinuum generation with low pump power.
High A_{eff} (Telecommunications Fibers): In long-haul optical communication systems, the goal is to maximize A_{eff} (using designs such as Large Mode Area—LMA—fibers) to mitigate nonlinear distortions that degrade the data signal.

III. WAVEGUIDE GEOMETRY

The waveguide modeled in this study has a height of 220 nm. This dimension is determined by the fabrication process to be used for its construction. The waveguide width suggested in the task description is 500 nm. The decision to maintain this suggested width simplifies the work . Figure 1 shows the waveguide geometry configuration end Figure 2 shows the waveguide layout at Lumerical MODE software.

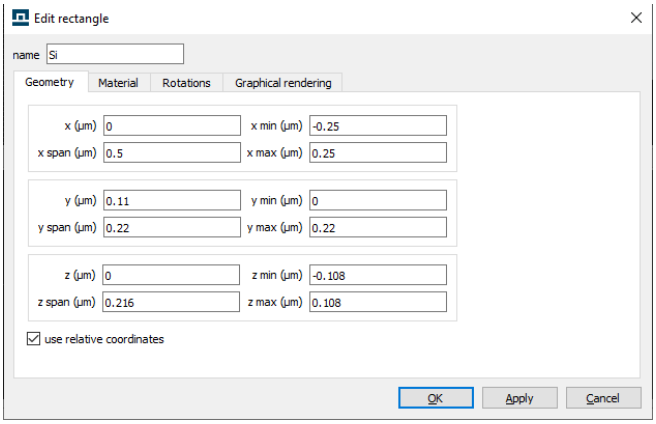


Figure 1 - Wave Guide Geometry

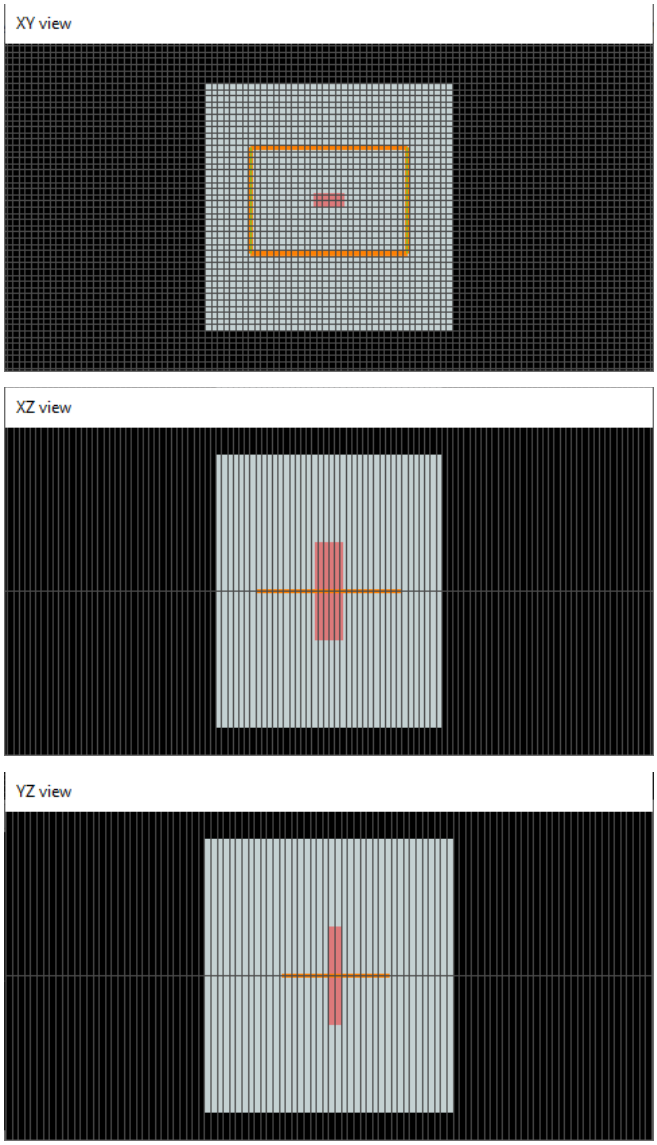


Figure 2 – XY , XZ, end YZ waveguide views

IV. MODAL ANALYSIS

Modal analysis was made by FDE Eingensolver using Lumerical Mode tools. Table 1 shows the four firsts propagation modes of the waveguide fot a wavelenght of 1550 nm, and Figure 3 shows Fundamental Mode E intensity.

Table 1

mode #	effective index	wavelength (μm)	loss (dB/cm)	group index	TE polarization fraction (Ex)	waveguide TE/TM fraction (%)	effective area (μm^2)
1	2.447629+1.208871e-09i	1.55	0.00042564	4.197807+2.902881e-09i	98	76.18 / 81.96	0.190486
2	1.770306+6.643812e-10i	1.55	0.00023393	3.699583+5.064959e-09i	4	68.94 / 88.9	0.368746
3	1.497693+3.497787e-10i	1.55	0.00012316	2.446871+3.834211e-09i	65	86.47 / 95.99	0.825281
4	1.348011+3.904566e-11i	1.55	1.3748e-05	1.628890+6.020957e-11i	99	96.5 / 91.35	2.24944

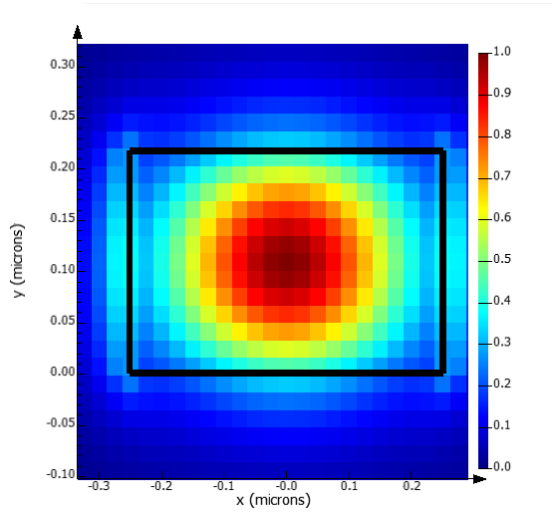


Figure 3 - Fundamental model E Intensity

It is also possible to determine some waveguide parameters using the frequency sweep function. The following are graphs representing the parameters n_{eff} (Figure 4), n_g (Figure 5), progapation index β (Figure 6), waveguide loss (Figure 7) and gain (Figure 8).

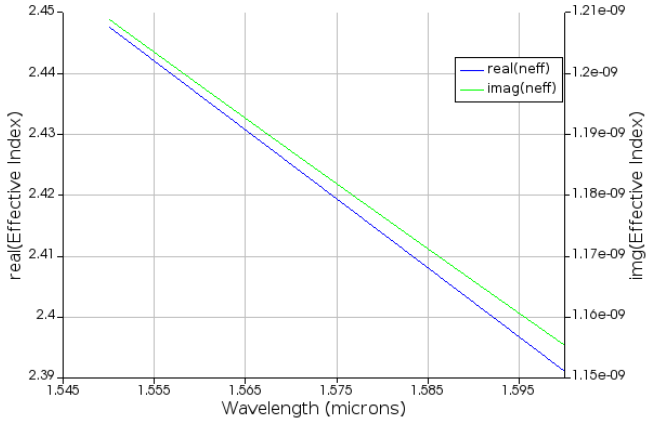


Figure 4 - Wave guide n_{eff} index.

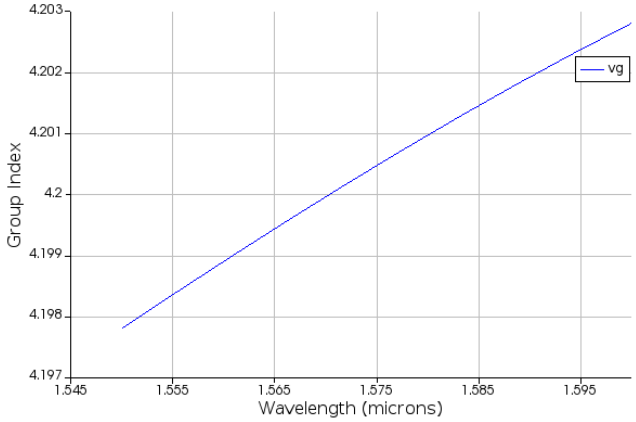


Figure 5 - Wave guide group index vs. wavelength

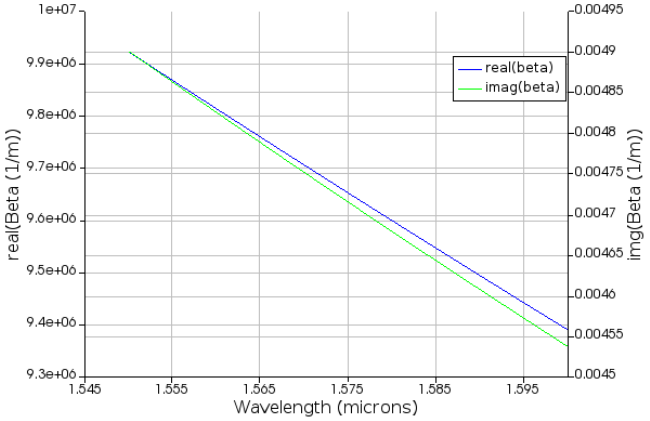


Figure 6 - Wave guide Propagation Index vs. Wavelength

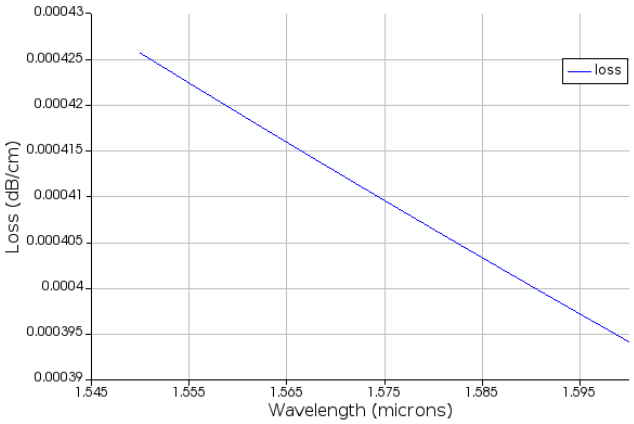


Figure 7 - Waveguide loss vs. Wavelength

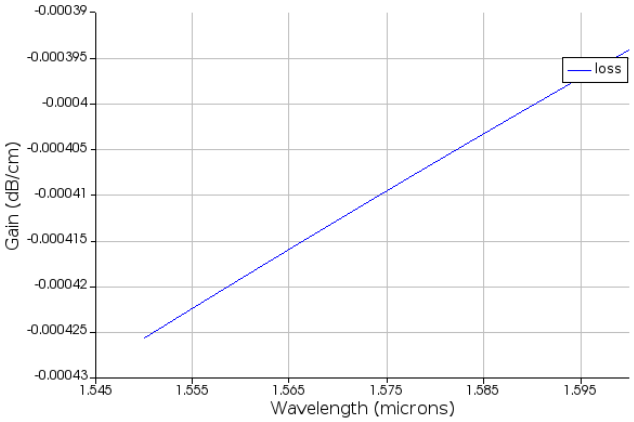


Figure 8 - Waveguide Gain vs, Wavelength

V. WAVEGUIDE COMPACT MODELO

Other work aim is to extract compact model for the selected wave guide. To achieve this objective, we used the script provided by the course organizers. To run it, we selected it in the Script File Editor window and executed it while keeping the waveguide simulation file open in Lumerical MODE. The result obtained is presented below. (2.44763, -1.12923, -0.0319043) these values are transported for the polynomial equation.

$2.44763, -1.12923(\lambda - \lambda_0), -0.0319043(\lambda - \lambda_0)^2$

The script used to extract the polynomial parameters (`compact_extract.lsf`) is also used to generate a plot of the fitted n_{eff} curve. The generated plot is shown in the Figure 9.

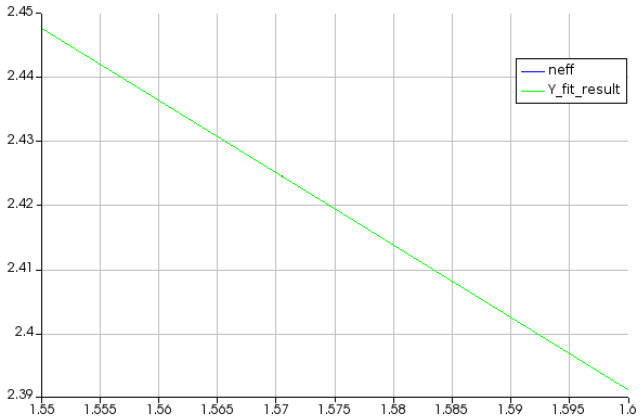


Figure 9 - Y-fit plot

VI. THE MACH–ZEHNDER INTERFEROMETER

A Mach-Zehnder interferometer (MZI) is an interferometric device that makes use of two interfering paths of different lengths to resolve different wavelengths. Today, Mach-Zehnder interferometers are typically constructed in integrated optics and consist of two 3 dB directional couplers interconnected through two paths of differing lengths, as shown in Figure 3.21(a). The substrate is usually silicon, and the waveguide and cladding regions are silica (SiO2). [1]

A Mach-Zehnder interferometer (MZI) is an interferometric device that makes use of two interfering paths of different lengths to resolve different wavelengths. Devices constructed on this principle have been around for some decades. Today, Mach-Zehnder interferometers are typically constructed in integrated optics and consist of two 3 dB directional couplers interconnected through two paths of differing lengths. The substrate is usually silicon, and the waveguide and cladding regions are silica (SiO2), as shown in Figure 10. [2]

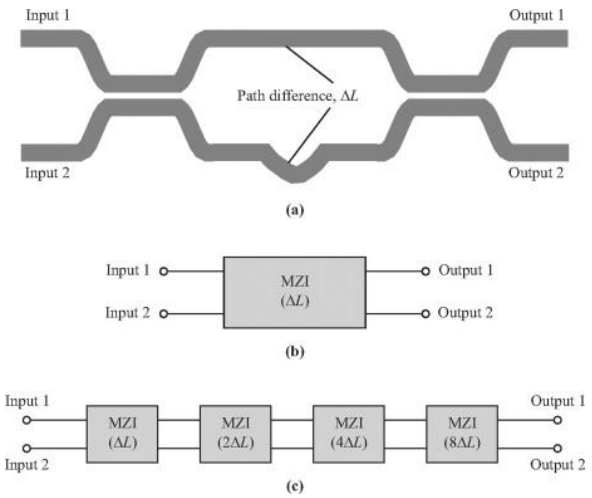


Figure 10 - (a) An MZI constructed by interconnecting two 3 dB directional couplers. (b) A block diagram representation of the MZI in (a). ΔL denotes the path difference between the two arms. (c) A block diagram of a four-stage Mach-Zehnder interferometer, which

Mach-Zehnder interferometers are useful as both filters and (de)multiplexers. Even though there are better technologies for making narrow band filters, for example, dielectric multicavity thin-film filters, MZIs are still useful in realizing wide band filters. For example, MZIs can be used to separate the wavelengths in the 1.3 μm and 1.55 μm bands. Narrow band MZI filters are fabricated by cascading several stages, as we will see, and this leads to larger losses. In principle, very good crosstalk performance can be achieved using MZIs if the wavelengths are spaced such that the undesired wavelengths occur at, or close to, the nulls of the power transfer function. However, in practice, the wavelengths cannot be fixed precisely (for example, the wavelengths drift because of temperature variations or age). Moreover, the coupling ratio of the directional couplers is not 50:50 and could be wavelength dependent. As a result, the crosstalk performance is far from the ideal situation. Also, the passband of narrow band MZIs is not flat. In contrast, the dielectric multicavity thin-film filters can have flat passbands and good stop bands.

MZIs are useful as two-input, two-output multiplexers and demultiplexers. They can also be used as tunable filters, where the tuning is achieved by varying the temperature of one of the arms of the device. This causes the refractive index of that arm to change, which in turn affects the phase relationship between the two arms and causes a different wavelength to be coupled out. The tuning time required is of the order of several milliseconds. For higher channel-count multiplexers and demultiplexers, better technologies are available today. One example is the arrayed waveguide grating (AWG) described in the next section. Since understanding the MZI is essential to understanding the AWG, we will now describe the principle of operation of MZIs.

A. Principle of Operation

Consider the operation of the MZI as a demultiplexer; so only one input, say, input 1, has a signal as Figure 10 . After the first directional coupler, the signal power is divided equally between the two arms of the MZI, but the signal in one arm has a phase shift of $\pi/2$ with respect to the other. Specifically, the signal in the lower arm lags the one in the upper arm in phase by $\pi/2$. Since there is a length difference of ΔL between the two arms, there is a further phase lag of $\beta \Delta L$ introduced in the signal in the lower arm. In the second directional coupler, the signal from the lower arm undergoes another phase delay of $\pi/2$ in going to the first output relative to the signal from the upper arm. Thus, the total relative phase difference at the first or upper output between the two signals is $\pi/2 + \beta \Delta L + \pi/2$. At the output directional coupler, in going to the second output, the signal from the upper arm lags the signal from the lower arm in phase by $\pi/2$. Thus, the total relative phase difference at the second or lower output between the two signals is $\pi/2 + \beta \Delta L - \pi/2 = \beta \Delta L$.

If $\beta \Delta L = k\pi$ and k is odd, the signals at the first output add in phase, whereas the signals at the second output add with opposite phases and thus cancel each other. Thus, the wavelengths passed from the first input to the first output are those wavelengths for which $\beta \Delta L = k\pi$ and k is odd. The wavelengths passed from the first input to the second output are those wavelengths for which $\beta \Delta L = k\pi$ and k is

even. This could have been easily deduced from the transfer function of the MZI in the following equation (6), but this detailed explanation will help in the understanding of the arrayed waveguide grating.

Assume that the difference between these path lengths is ΔL and that only one input, say, input 1, is active. Then it can be shown that the power transfer function of the Mach-Zehnder interferometer is given by

$$\begin{pmatrix} T_{11}(f) \\ T_{12}(f) \end{pmatrix} = \begin{pmatrix} sin^2(\frac{\beta \Delta L}{2}) \\ cos^2(\frac{\beta \Delta L}{2}) \end{pmatrix} \quad (6)$$

Thus, the path difference between the two arms, ΔL, is the key parameter characterizing the transfer function of the MZI.

In its classic and ideal form (using two perfect 50:50 optical couplers/beam splitters), the optical power transfer function to one of the output ports in terms of the wavelength (λ) is given by:

$$T(\lambda) = \frac{I_{saída}}{I_{entrada}} = \cos^2\left(\frac{\pi \cdot n_{eff} \cdot \Delta L}{\lambda}\right) \quad (7)$$

VII. Mach-Zehnder Interferometers Simulations

The initial simulation performed for the Mach-Zehnder interferometer aimed to evaluate the impact of variations in waveguide length difference on power transmission at the device outputs. The Figure 11 shows the simulation circuit implemented in Lumerical Interconnect.

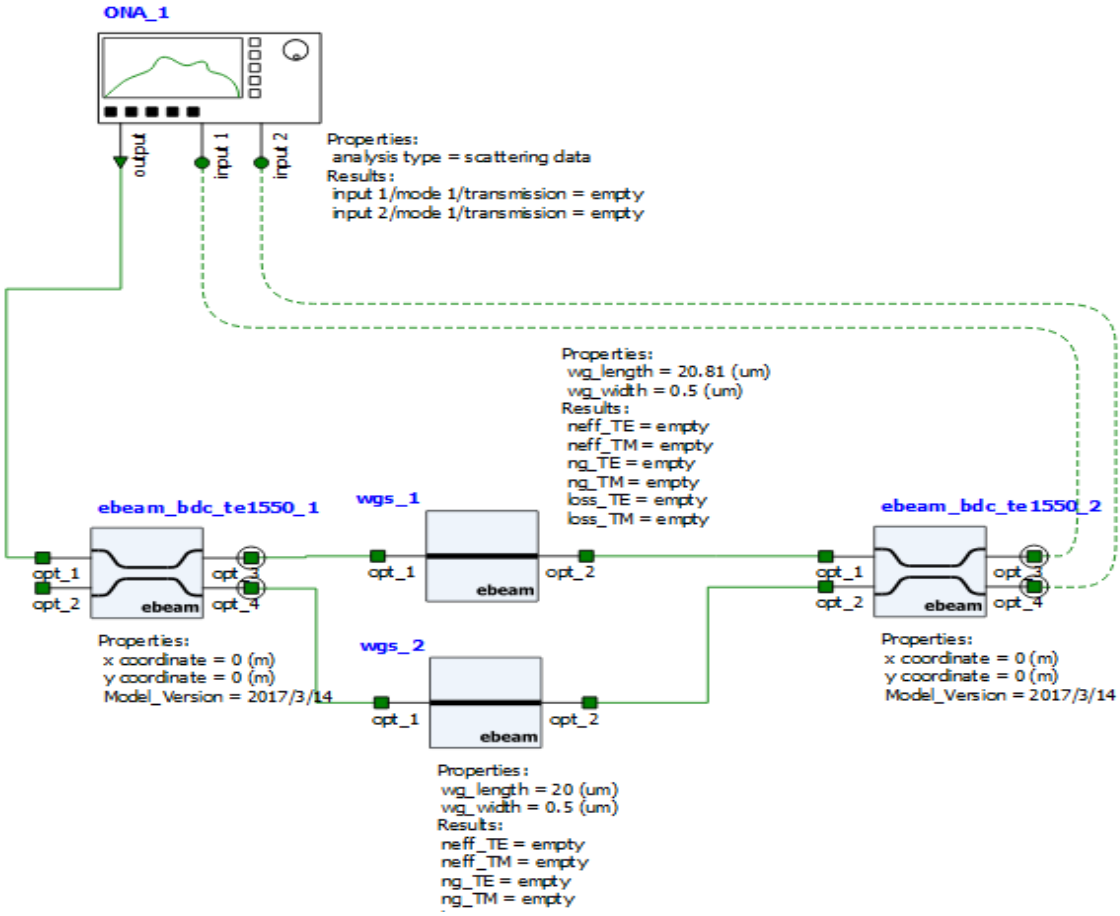


Figure 11 - Mach-Zehnder interferometer Simulation Circuit

The first simulation was performed with both waveguides having the same geometric dimensions, with the aim of verifying the impact of varying the waveguide length on the field at the two output ports. Figure 12 Shows the results.

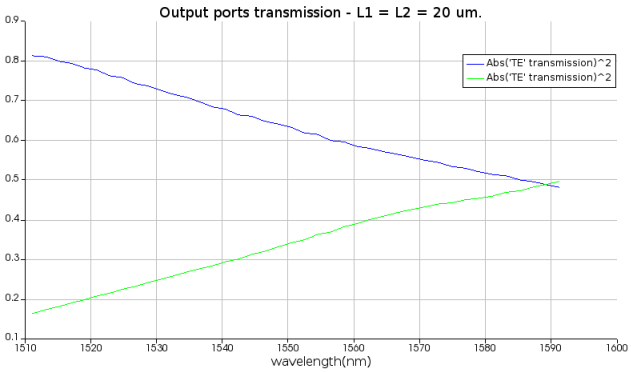


Figure 12 - Transmission characteristics - L1 = L2 = 20 um

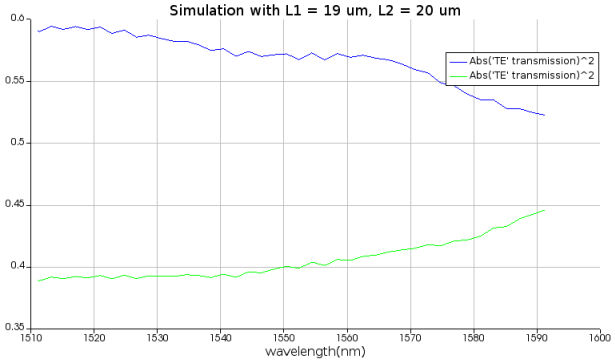


Figure 13 - Transmission characteristics L = 19 um

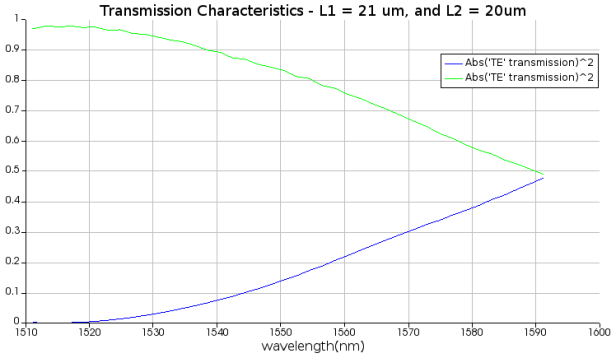


Figure 14 - Transmission characteristics L = 21 um

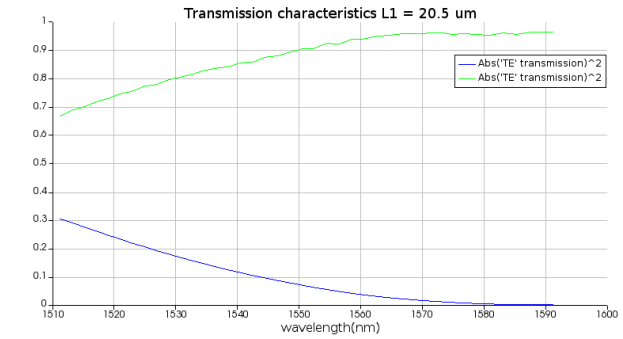


Figure 15 - Transmission characteristics L = 20.5 um

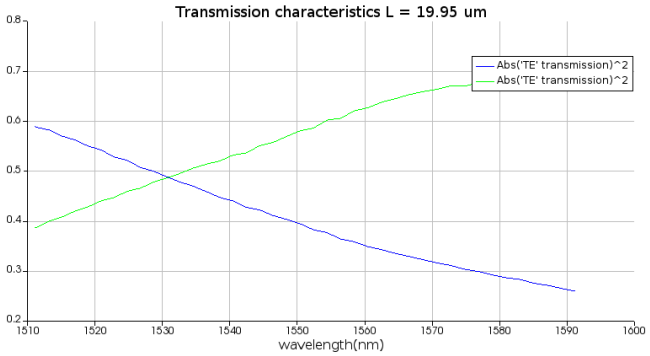


Figure 16 - Transmission characteristics L = 19.95 um

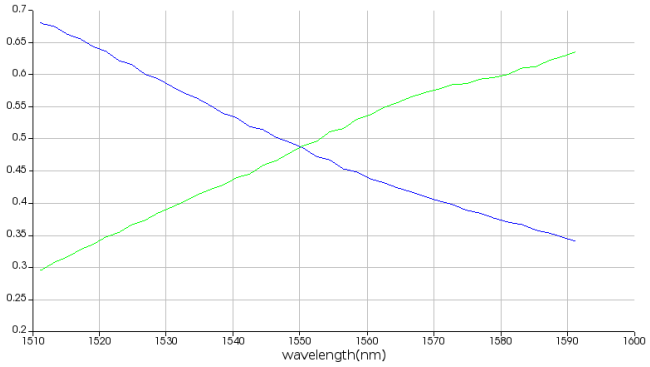


Figure 17 - Transmission characteristics L1 = 19.969 um

The series of simulations performed—partially shown in Figures 12 through 17—illustrates how transmission varies with changes in ΔL. By configuring both waveguides to the same length, it was possible to identify the point where transmission at the two output ports is balanced at approximately 50% for each port. A target of achieving a 50%-50% split was set; by manually varying the values of L1 and observing the separation and convergence of the curves (as shown in Figure 19), the expected result was obtained—although the intersection point of the two curves occurred at a wavelength of 1549.52 nm, with a transmission rate of 48.43% at each port. This leads to the conclusion that there is a 3.16% loss in the simulated circuit.

It is worth noting that the simulations allow for obtaining the waveguide values, as detailed in the Figure 18.

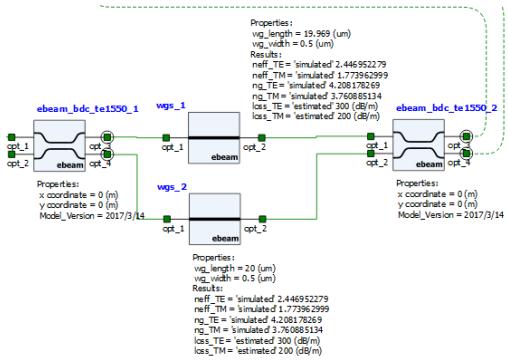


Figure 18 - Details of the parameters obtained in the simulation

VIII. Final Project

The final project was done with four MZI circuits. The cells list is shown in Figure 19.

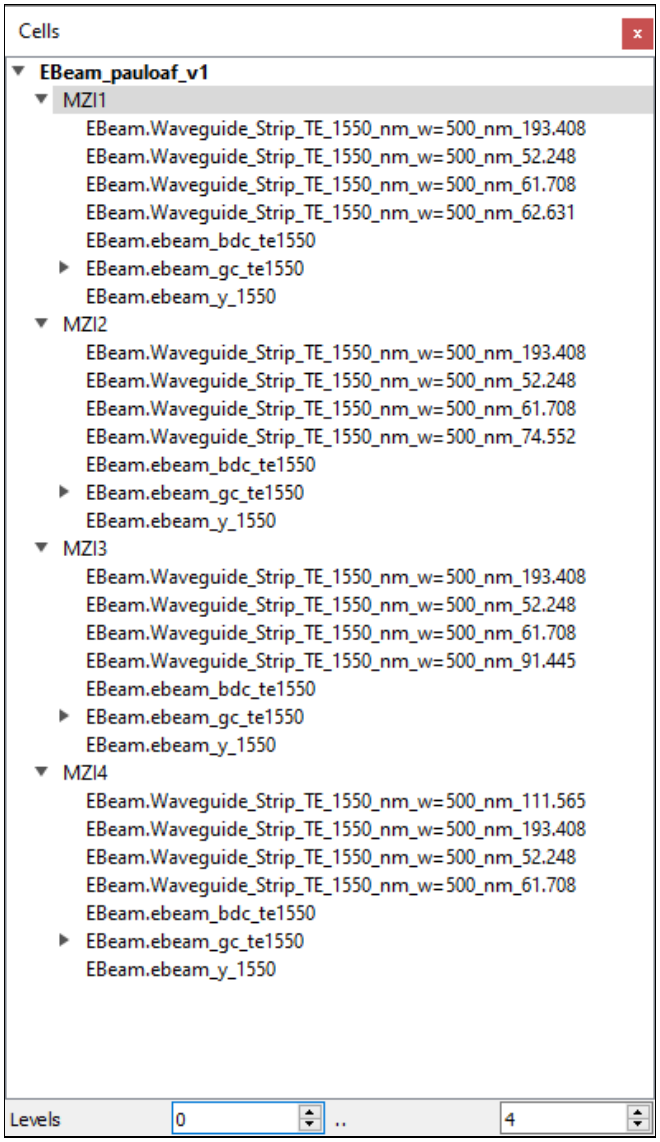


Figure 19 - Final project cells list

The waveguides length differences were measured, and the results is shown below:

MZI1 – 10.383 um
MZI2 – 22.304 um
MZI3 – 39.197 um
MZI4 – 59.317 um

The SiEPIC KLayout Design Rule Check (DRC) EBeam Verification was executed, and final report is shown in Figure 20

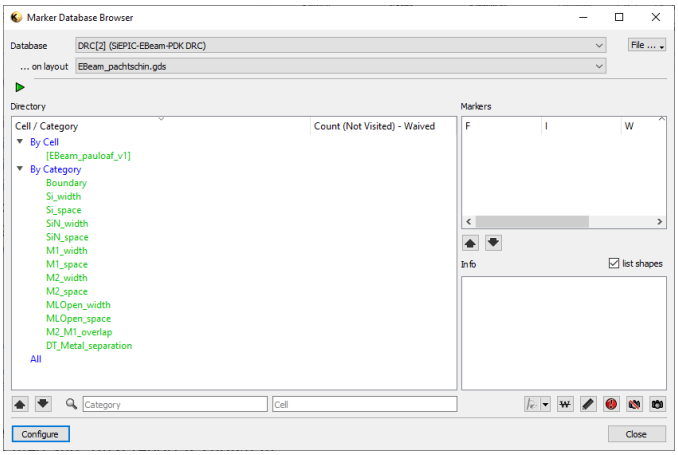


Figure 20 - Final Design DRC SiEPIC Verification.

Functional Layout Check was executed, and the final message was “No layout errors detected”.

Automated extraction of measurement coordinates was performed, and the four laser inputs were correctly registered, as shown in Figure 21.

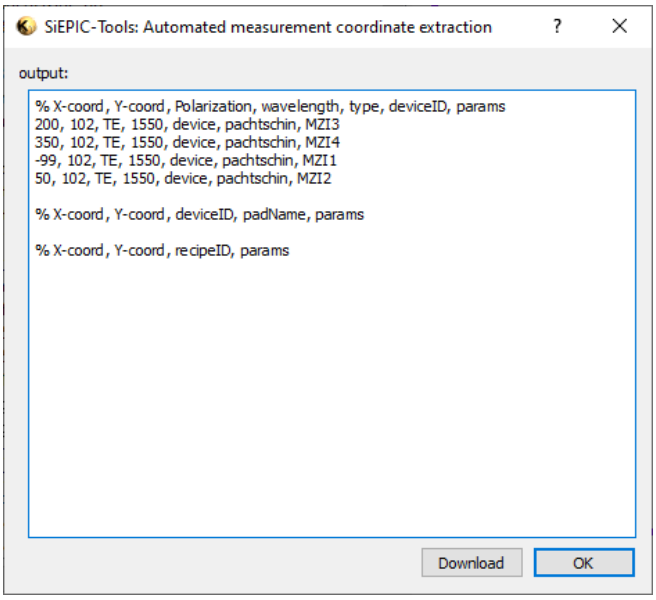


Figure 21 - Automated measurement coordinate extraction result.

The Interconnect simulations were done for each one of the MZI circuits. Figure 22 shows the MZI1 Interconnect circuit simulation. The gain at input ports one and two of MZI1, MZI2, MZI3 and MZI4 are shown in Figure 23, Figure 24, Figure 25 end Figure 26 respectively.

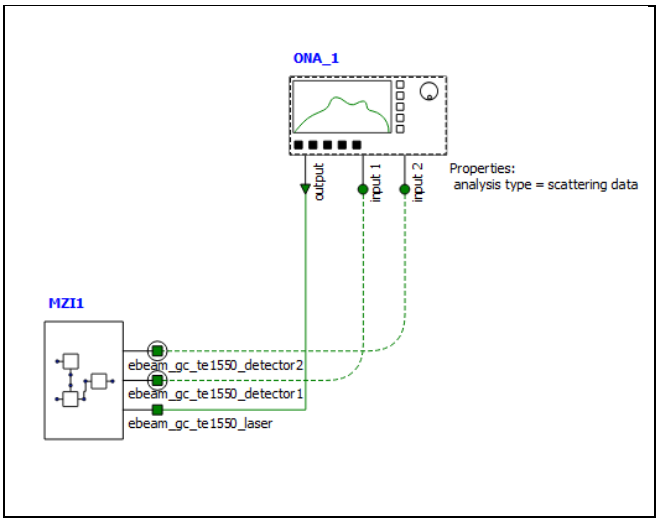


Figure 22 - MZI1 Interconnect circuit simulation

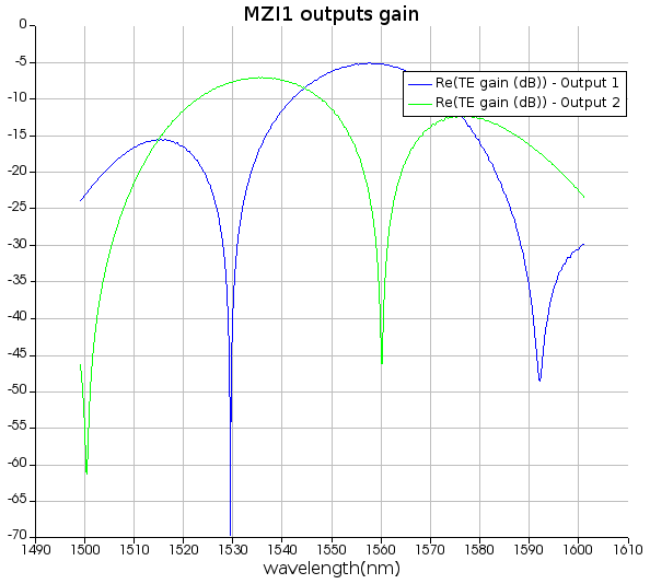


Figure 23 - MZI1 outputs gain

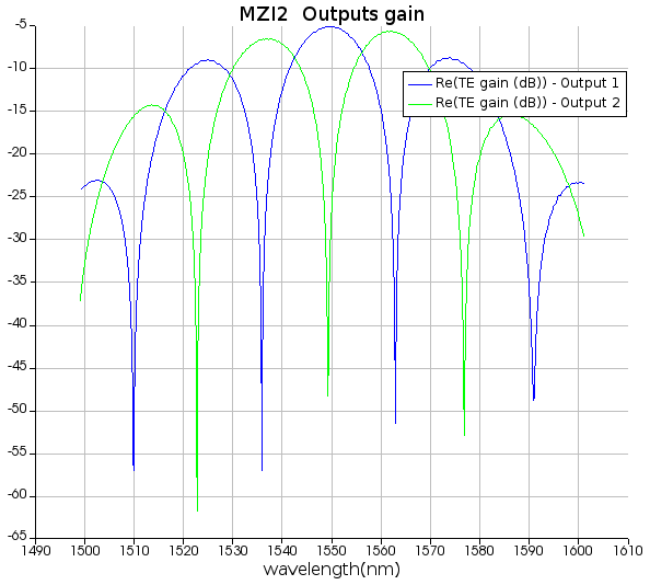


Figure 24 - MZI2 Outputs gain

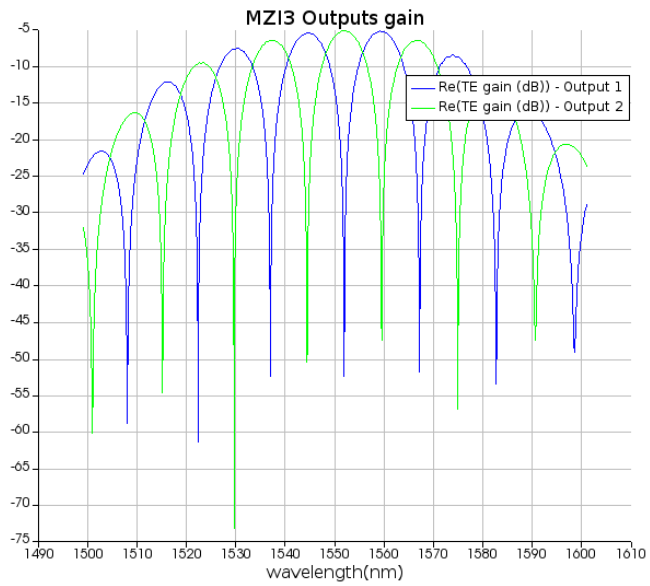


Figure 25 - MZI3 Outputs gain

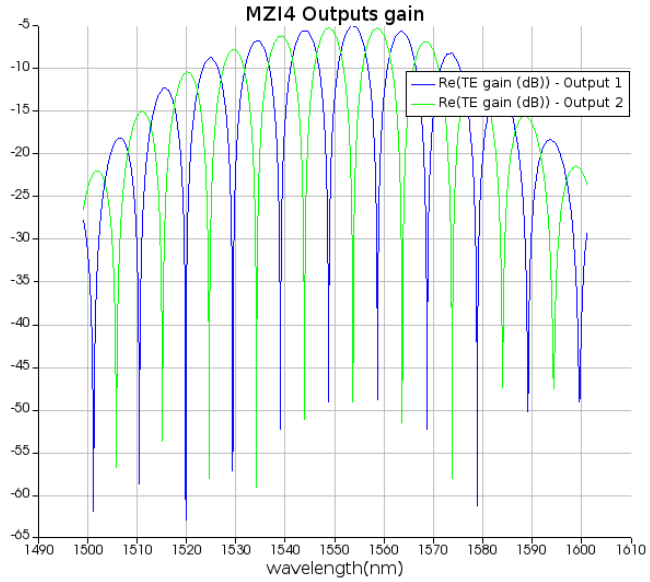


Figure 26 - MZI4 outputs gain

IX. CONCLUSIONS

This work demonstrated how waveguide parameters can be obtained using Lumerical's MODE and Interconnect modules. Many other features are available within the Lumerical tool suite, and results can be compared using methods such as FDTD or FDE, bearing in mind that each tool is best suited for a specific application.

After the simulations the circuit layout was designed using Klayout tool and verifications were made showing that the layout is good enough to be fabricated.

Interconnect simulations were done and the results shows the FSR variations as the waveguide's length differences change.

The Final Layout is Shown in Figure 27.

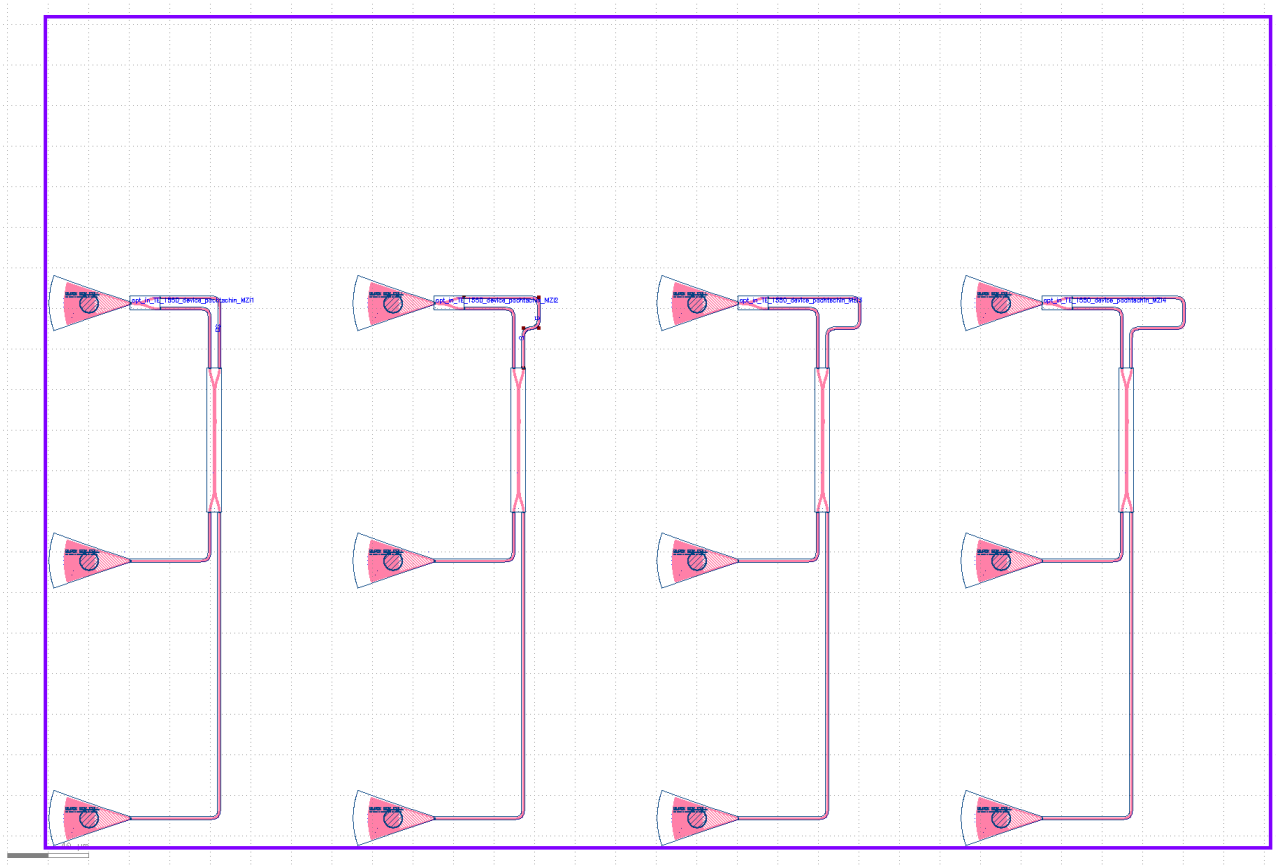


Figure 27 - Final Layout

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